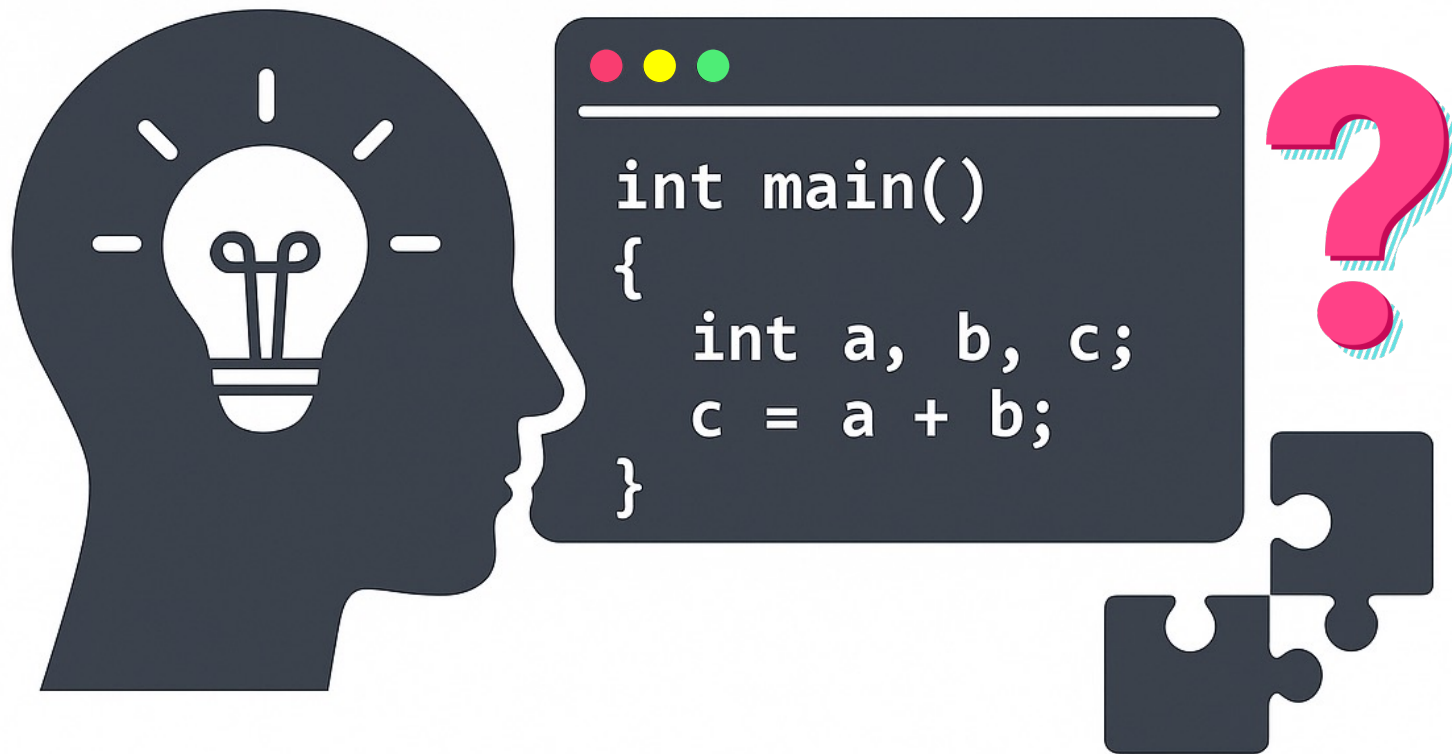


PROBLEM SOLVING - 1



1	Concatenation of String
Problem Statement	Write a program to read a word from the console, concatenate " a student " to it and print it back to the console.
Input Format	The only line of input has a string.
Output Format	A single string.
Sample Input	Quick
Sample Output	Quick a student

2	Add an Integer
Problem Statement	Read a number from the console, add 3 to it and print it.
Input Format	The first line contains the number.
Output Format	A single integer.
Sample Input	2
Sample Output	5

3	Name Age Roll Number
Problem Statement	Read name, age and roll number from three different lines and print them back to the console, in a single line.
Input Format	Three strings in three separate lines.
Output Format	A string in a single line.
Sample Input	Ram 32 4567
Sample Output	Ram 32 4567

4	Sum of integers
Problem Statement	Read four integers from the console and print their sum. The input will be a single line containing the four integers, separated by single whitespaces.
Input Format	Four integers in a single line separated by space.
Output Format	A single integer.
Sample Input	2 8 9 12
Sample Output	31

5	Product of integers
Problem Statement	Read three integers from the console and print their product.
Input Format	The input will be a single line containing the three integers, separated by single whitespaces.
Output Format	A single integer.
Sample Input	23 10 10
Sample Output	2300

6

Float divided by integer

Problem
Statement

Read a floating-point number from the console, divide it by 2, and print it rounded to 2 decimal place.

Input
Format

A single floating-point number.

Output
Format

A single integer.

Sample
Input

2.5

Sample
Output

1.25

7	Greater Average
Problem Statement	You are given 3 numbers A, B, and C. Determine whether the average of A and B is strictly greater than C or not? NOTE: Average of A and B is defined as $(A+B)/2$. For example, average of 5 and 9 is 7, average of 5 and 8 is 6.5.
Input Format	Single line of input contains three space separated integers.
Output Format	"YES" or "NO" without quotes.
Sample Input	5 9 6
Sample Output	YES

8

Valid Phone Number

Problem Statement

In PhoneLand, a valid phone number consists of 5 digits with no leading zeros.

For example, 98765, 10000, and 71023 are valid phone numbers, while 04123, 9231, and 872310 are not.

Alex went to a store and purchased N items, where the cost of each item is X . Determine whether the total bill is equivalent to a valid phone number.

Input Format

The first line of input will consists of two space-separated integers N and X — the number of items Alex bought and the cost per item.

Output Format

For each test case, output on a new line, YES, if the total bill is equivalent to a valid phone number and NO otherwise.

Sample Input

25 785

Sample Output

YES

9

String ValidationProblem
Statement

Given a string S, determine if it contains both the characters '&' and '#' and the length of the string is even. If both conditions are satisfied, print 'YES'; otherwise, print 'NO'.

Input
Format

Single line input string

Output
Format

"YES" or "NO".

Sample
Input

abc&def#

Sample
Output

YES

10	Adding first-2 and last element-2 of list
Problem Statement	Given N integers. Find the sum of the first 2 elements and last 2 elements in separate lines. You can assume that $N \geq 4$.
Input Format	First line contains N. The next line contains the N space separated integers.
Output Format	Two integer in two different line.
Sample Input	18 1 2 3 4 5 6 7 8 9 10 12 13 15 21 22 45 90 100
Sample Output	3 190

11

Count and Compare Odds and EvensProblem
Statement

Given a list of N positive integers. You need to find out the count of odd and even integers. If the count of odd integers are greater than the count of even integers, print "Odd" (without quotes). If the count of even integers are greater than the count of odd integers, print "Even" (without quotes). Print "Tie" (without quotes) if count of odd and even integers are equal.

Input
Format

Two lines of input. First line contains an integer N, representing the total number of integers. Second line of input contains N space separated integers.

Output
Format

Single string in single line.

Sample
Input

5
10 24 3 37 38

Sample
Output

Even

12	Group Salary
Problem Statement	You are given salaries of N groups of employees (each group has M employees) in the format specified below. The count of employees in each group may not be the same. Print the sum of salaries of each group of employees.
Input Format	First line contains the value of N, followed by N pairs of lines. The first line of each pair contains the value of M and the second line of each pair contains space separated M integers, denoting the salary of each employees.
Output Format	N lines, each having single integer.
Sample Input	3 4 12000 10000 5000 8000 2 3000 2000 3 7000 9000 5000
Sample Output	35000 5000 21000

13	Group-wise Marks Analysis
Problem Statement	You are given marks of N groups of students in the format specified below. The strength of each group may not be the same. Print the maximum, minimum, total, and average marks (rounded upto one place of decimal) of each group of students.
Input Format	First line contains the value of N, followed by N pairs of lines. The first line of each pair contains the value of M and the second line of each pair contains space separated M integers, denoting the marks of each student.
Output Format	N lines, each having M space separated values.
Sample Input	3 3 12 14 16 4 11 13 18 19 6 12 14 16 18 20 13
Sample Output	16 12 42 14.0 19 11 61 15.2 20 12 93 15.5

14

Marks Sum of Top 5 and Last 5Problem
Statement

You are given marks of some students. Print the sum of the marks of the top 5 students, and the marks of last 5 students in decreasing order.

Input
Format

The first line contains the value of N and the next line has N space separated integers.

Output
Format

First line containing an integer value and second line contains 5 integer numbers.

Sample
Input

8
50 30 60 55 75 85 65 45

Sample
Output

340
60 55 50 45 30

15	Swap K-Blocks
Problem Statement	Given an array of N integers and a positive integer K. Swap every K element with the next K elements. N is a multiple of 2K. Use two nested loops.
Input Format	First line of input contains the two integers denoting the value of N and K. The second line contains N space separated integers.
Output Format	N space separated integers in a single line.
Sample Input	12 3 1 2 3 4 5 6 7 8 9 10 11 12
Sample Output	4 5 6 1 2 3 10 11 12 7 8 9

16

Remove ElementsProblem
Statement

Given an integer array having N elements. Find the number of occurrences of 1, and remove the element at the index, which is equal to the number of occurrences of 1. Print the modified array. Given that, count of 1 < N.

Input
Format

First line contains the value of N. The next line contains N space-separated integers.

Output
Format

N - 1 space-separated integers.

Sample
Input

9
2 4 1 6 1 8 4 1 5

Sample
Output

2 4 1 1 8 4 1 5

17

Split The ArrayProblem
Statement

Given an integer array having N elements. Find the first locations of the minimum and maximum elements. Split the array into three parts from these locations. Build another array by concatenating the second part, followed by the first part, followed by the third part. Print the elements of this array in a single line.

It can be assumed that the minimum and maximum elements will not be the same.

Input
Format

First line contains the value of N. The next line contains N space-separated integers.

Output
Format

N space-separated integers.

Sample
Input

8
2 4 1 3 5 9 7 6

Sample
Output

1 3 5 2 4 9 7 6

18

Compare ArraysProblem
Statement

Given two arrays (A and B) having lengths N and M respectively. For each $B[i]$ print the count of elements from A, which is greater than $B[i]$.

Input
Format

First line has two space separated integers denoting N and M. The second line has N space separated integers. The third line has M space separated integers.

Output
Format

N space separated integers in a single line.

Sample
Input

```
5 7
2 6 1 3 9
5 1 8 7 0 4 12
```

Sample
Output

```
2 4 1 1 5 2 0
```

19

All Pairs with Sum KProblem
Statement

Given an array having N integers and an integer K. Find all pairs of elements from this array, such that their sum is K (brute force version of Two Sum problem). Also discuss the complexity of your approach.

Input
Format

First line of input contains the value of N and K. The second line contains N space separated integers.

Output
Format

Space separated elements of the pair. Each pair should go into a separate line.

Sample
Input

7 5
8 -1 3 4 -3 2 1

Sample
Output

8 -3
3 2
4 1

20

Next Greater Right ElementProblem
Statement

Given an array of N positive integers. For each element, print the first greater element towards the right of the element. If no greater element is found, print 0.

Input
Format

First line of input has N and the next line has N space separated integers.

Output
Format

N space separated integers in a single line.

Sample
Input

5
4 2 6 4 5

Sample
Output

6 6 0 5 0

21

Maximum ProfitProblem
Statement

Given an array, P, holding N integers, where P[i] represents the price of a particular stock on ith day. You are allowed to purchase and sell the stock only once. Find out your maximum profit.

Input
Format

First line contains the value of N. The next line has N space separated integers.

Output
Format

A single integer in a single line representing the maximum profit.

Sample
Input

5
1 5 2 3 1

Sample
Output

4

22

Maximum Subarray SumProblem
Statement

Given an array with N elements. Print the sum of the subarray having maximum sum (brute force version of Kadane Algorithm). The elements of the array could be positive, negative or zero. Also discuss the complexity of your approach.

Input
Format

First line contains the value of N. The next line has N space separated integers.

Output
Format

A single integer in a single line.

Sample
Input

7
8 -1 3 4 -3 2 1

Sample
Output

14

23

Row-wise Sum, Max, and MinProblem
Statement

An input is given in the following format. The first line has two space separated integers N and M, denoting the number of rows and columns of a matrix. Next N lines have M space-separated integers. Input should be taken as a list of lists. Print the sum, maximum, and minimum of each row, in a single line, separated by space.

Input
Format

First line has 2 space separated integers, denoting N and M. Next N lines have M space separated integers.

Output
Format

N lines, each having M values.

Sample
Input

```
3 5
12 4 7 9 2
1 5 8 3 13
2 42 1 6 9
```

Sample
Output

```
34 12 2
30 13 1
60 42 1
```

24

SubmatrixProblem
Statement

Given a matrix with N rows and M columns ($N, M \geq 6$). Print a submatrix obtained from mat after removing the first row, first column, last two rows, and last two columns. Use nested loops.

Input
Format

First row consists of N and M. Next N rows contain M space separated elements. Read the elements as matrix (mat).

Output
Format

N - 3 rows of integers, each having M - 2 elements. See sample below.

Sample
Input

```
6 7
1 1 1 1 1 1 1
2 2 2 2 2 2 2
3 3 3 3 3 3 3
4 4 4 4 4 4 4
5 5 5 5 5 5 5
6 6 6 6 6 6 6
```

Sample
Output

```
2 2 2 2
3 3 3 3
4 4 4 4
```

25

Swap The Numbers

Problem
Statement

You are given a matrix with N rows and M columns. Swap the last K elements from the first row with the first K elements from the last row. Swap last K elements from the second row and first K elements from the second last row. And so on. And print the modified matrix. You can assume that N is even and $K < M$. See sample below.

Input
Format

First line contains space separated values of N, M and K. Next N lines have M space separated integers.

Output
Format

N lines of output, each having M space separated elements.

Sample
Input

```
4 5 2
1 2 3 4 5
5 6 7 8 9
9 10 11 12 13
13 14 15 16 17
```

Sample
Output

```
1 2 3 13 14
5 6 7 9 10
8 9 11 12 13
4 5 15 16 17
```

26

Swap Matrix ElementsProblem
Statement

You are given a matrix with N rows and M columns. Swap each element with the element in its immediate bottom-right cell (if the cell exists). Print the modified matrix.

Input
Format

First line of input contains the two integers denoting the values of N and M. The next N lines contain M space separated integers.

Output
Format

N lines, each with M space separated integers.

Sample
Input

```
3 4
24 29 12 33
25 36 18 17
31 22 43 67
```

Sample
Output

```
36 18 17 33
22 43 67 12
31 25 24 29
```

27

Insert String at Intervals

Problem Statement

You are given two strings S1 and S2 and a number N. Insert S2 in S1 after every N characters (see sample input). Print the modified String. Assume $0 < N < \text{len}(S1)$

Input Format

First two line has input string and third line has a input number.

Output Format

Single input string.

Sample Input

quick fox

2

Sample Output

qu#ic#k #fo#x

28

ASCII Range Character CounterProblem
Statement

Given a string (S) of length K, and two integers, N and M. Count the number of characters, whose ASCII value is within the range defined by N, M (both inclusive). Given that $N < M$.

Input
Format

First line has 3 space separated integers K, N and M. Next line contains a string.

Output
Format

A single input.

Sample
Input

15 67 75
AB*abXYDEF#@pqr

Sample
Output

3

29

Strip CharactersProblem
Statement

Given a string (S) of length N, and an integer (K). Print S, after stripping the first K and the last K characters. You can assume that $2K < N$.

Input
Format

First line has two input numbers N and K. Second line has a input string.

Output
Format

Single line string output.

Sample
Input

11 3
abcdefghijkl

Sample
Output

defgh

30

Rotate String - 1Problem
Statement

Rotate the given string S of length N by K places towards left.
Given that $K \leq N$.

Input
Format

First line has 2 space separated number N and K.
Second line has an input string.

Output
Format

A single line output string.

Sample
Input

13 3
quickfoxbrown

Sample
Output

ckfoxbrownqui

31

Rotate String - 2Problem
Statement

Rotate the given string S of length N by K places towards right.

Input
Format

First line has 2 space separated number N and K.
Second line has an input string.

Output
Format

A single line output string.

Sample
Input

13 3
quickfoxbrown

Sample
Output

ownquickfoxbr

32

Check PalindromeProblem
Statement

Given a string (S) of length N. Check whether it is a palindrome or not.

Input
Format

First line has a single input number N. Second line has an input string.

Output
Format

'True' or 'False', without the quotes.

Sample
Input

10
quickquick

Sample
Output

False

33

Reverse K-ChunkProblem
Statement

You are given a string (S) of length N, and an integer (K). Create a new string in the following fashion.

- 1) Pick first K characters as it is.
- 2) Followed by next K characters in reverse order.
- 3) Followed by next K characters as it is and so on.

Use slicing. Do not use nested loops. You can assume that N is a multiple of 2K.

Input
Format

First line has 2 space separated number N and K.
Second line has an input string.

Output
Format

A single line output string.

Sample
Input

12 3
abcdefghijkl

Sample
Output

abcfedghilkj

34

Rotate and Check PalindromeProblem
Statement

Given two strings, S1 and S2, and a positive integer K.

- 1) Rotate S1 rightwards by K positions and S2 leftwards by K positions.
- 2) Let S3 be the string obtained by combining the rotated version of S1 and S2.
- 3) If S3 is a palindrome print S3 followed by "True" otherwise print S3 followed by "False", without quotes.

Input
Format

First line has three space separated integers , length of first two strings and the value of K. Next two line has an input strings.

Output
Format

Output string followed by True or False.

Sample
Input

5 4 3
abcde
abcd

Sample
Output

cdeabdabc False

35	Transform String
Problem Statement	Given a string consisting of digits, alphabets and special characters. Transform this string into another by converting all alphabets into small case letters, leaving digits as it is and moving the special characters at the end. Print the transformed string.
Input Format	First line has a single input number N. Second line has an input string.
Output Format	A single line output.
Sample Input	15 abc12AD34@##123
Sample Output	abc12ad34123@##

36	K Rotate
Problem Statement	Given an alphanumeric string, S. Let the sum of squares of digits in S be K. If K is even, rotate the alphabets of S rightwards by K positions, otherwise rotate it leftwards by K positions. Print the rotated string.
Input Format	First line has a single input number N. Second line has an input string.
Output Format	A single line output.
Sample Input	9 2ab3cd45e
Sample Output	bcdea

37

First & Last Word LengthsProblem
Statement

You are given N sentences, each having M number of words. The first line of input contains space separated values of N and M. Next N lines have M space-separated strings. Read this input as a list of list. Print the lengths of the first and the last words from each row, in a single line, separated by space. See sample below.

Input
Format

See problem statement.

Output
Format

See problem statement.

Sample
Input

3 4
quick brown fox jumps
alpha beta gamma zeta
quick brown alpha beta

Sample
Output

5 5
5 4
5 4

38

Count of InstancesProblem
Statement

You are given two strings (S1 and S2) having lengths N and M respectively. Print the count of instances of each character from S1 in S2.

Input
Format

First line has two space separated integers denoting N and M. The second and third lines of input contain strings representing S1 and S2.

Output
Format

N integers separated by space in a single line.

Sample
Input

19 24
The quick brown fox
jumps over the lazy dogs

Sample
Output

0 1 2 4 0 1 0 0 0 4 0 1 2 0 0 4 0 2 0

39

Characters from Paired StringsProblem
Statement

Given N (N is even) strings of equal length. Make a string by concatenating the first character of the first string and the last character of the last string. Make another string by concatenating the first two characters of the second string and the last two characters of the second last string. And so on. Print the strings so formed in the same line, separated by space. Use string slicing. You can assume that the length of any string $> N/2$.

Input
Format

First line has an even input number N. Second line has N space separated input strings.

Output
Format

N space separated strings.

Sample
Input

6
alpha kumar polio virus hello gamma

Sample
Output

aa kulo polrus

40

Character GapProblem
Statement

Given a string S of length N, consisting of lower case English alphabets. For each character in S, count the number of characters between the chosen character and its next occurrence. If no next occurrence is found print -1.

Input
Format

First line of input contains an integer denoting the value of N. The second line contains a string denoting S.

Output
Format

N space separated integers in a single line.

Sample
Input

7
ababaad

Sample
Output

1 1 1 -1 0 -1 -1